

Product Requirement Document (PRD)

Face Attendance System (Sentinel)

Course / Track: On-the-Job Training (OJT) Project Framework

Track: Multimodal AI (Computer Vision + Language / LLM)

Document: Comprehensive PRD v1.1 (16 chapters) - Resubmit after mentor review

Status: Addresses feedback from Soumen Mukherjee (2026-09-03)

Revision note: v1.0 incorrectly treated blink (a second visual cue) as the multimodal partner. v1.1 defines Vision + Language (LLM) as the two modalities, adds PostGIS as the geospatial layer, and expands to a full 16-chapter package with database schemas, API specifications, and a requirements traceability matrix.

Chapter 1 - Document Overview & Metadata

Field	Description / Student Inputs
Project Title	Face Attendance System (working name: Sentinel)
Project Track	Multimodal AI - Vision + Language (LLM)
Team Members & IDs	Pratyksh Gupta - MSU ID: 240410700133
Assigned Mentor	Soumen Mukherjee
Document Version & Date	v1.1 3 September 2026 (resubmit)
Primary deliverable	Deployed full-stack multimodal AI app: Web UI + REST API
Domain	Computer Vision + NLP/LLM - automate attendance & org Q&A
Reference papers	ArcFace (CVPR 2019); FaceNet (CVPR 2015); RAG (Lewis et al., 2020)
Eval dataset	LFW View 2 (face); org attendance corpus + WikiQA-style FAQ for RAG lab

Document map (16 chapters)

Ch.	Title
1	Document Overview & Metadata
2	Executive Summary & Problem Statement
3	Goals, Scope & Success Metrics
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Chapter 2 - Executive Summary & Problem Statement

2.1 Problem Statement

Schools, campuses, and workplaces still mark attendance with paper registers, shared PINs, RFID cards, or a remote “I am here” button. These methods are easy to share, easy to fake, and hard to audit. Separately, staff and managers ask the same questions every morning (“Who is late?”, “Is Rohan present?”, “Who still has no photo?”) and get answers only by scrolling spreadsheets.

This project builds a multimodal face attendance platform: a camera proves who is present (vision), and a natural-language assistant answers what the attendance record means (language / LLM). Check-in can also be locked to an organisation site using PostGIS geofences so a remote punch fails.

2.2 Why current solutions are insufficient

- Consumer face unlock is one device / one user - not a multi-person org gallery with HR dashboards.
- Pure CV demos rarely expose evaluation (TAR@FAR) or a language layer for operators.
- Spreadsheets answer questions slowly and do not ground answers in live attendance rows.
- Haversine-only geofence prototypes do not meet the reference requirement for PostGIS spatial types and queries.

Chapter 3 - Goals, Scope & Success Metrics

3.1 Primary goals

- Enroll faces; identify people at check-in with a published encoder (ArcFace / FaceNet).
- Fuse vision identity with optional blink anti-spoof (vision-internal), then write attendance.
- Provide a text / LLM assistant that answers attendance questions with grounded SQL / RAG retrieval.
- Enforce site presence with PostgreSQL + PostGIS (geography, ST_DWithin).
- Ship a React web app + FastAPI REST API with JWT roles (platform admin, org admin/HR, employee).

3.2 Success metrics

Metric	Target
Dashboard load	< 2 s on campus Wi-Fi
Authentication	JWT; hashed passwords; role-restricted routes
CRUD accuracy	Person create/update/deactivate visible on dashboard
Report generation	CSV export; LLM answers cite attendance facts
UI	Responsive desktop + phone browser
Recognition	TAR@FAR = 1e-3 on LFW View 2; threshold from that curve
Check-in latency	Detect + embed + match < 1 s CPU (one face)
Geofence	PostGIS ST_DWithin; outside radius → reject
LLM grounding	≥ 90% of lab FAQ answers cite a retrieved attendance / person row

3.3 In scope / out of scope

In scope (v1.1)	Out of scope
Org login, face enroll/check-in, Today dashboard, CSV, LLM chat, PostGIS geofence, blink anti-spoof	Payroll/leave, native mobile stores, training a face net from scratch on private mega-data, certified ISO PAD

Chapter 4 - Stakeholders, Personas & User Journeys

Persona	Needs
Platform admin	Create organisations; open any org; no org code at login
Org admin / HR	Enroll faces, set sites/geofence, Today board, ask LLM “who is late?”
Employee	Org code + login; camera check-in; see own history
Reception / operator	Kiosk check-in; verify faces in lab
Mentor / evaluator	Measurable multimodal fusion, PostGIS, full docs

4.1 Journey - face check-in (vision)

Employee enters org code → signs in → camera captures frame (+ blink burst if required) → GPS sent when geofence active → PostGIS tests point-in-radius → face embed + gallery match → present row written → Today board updates.

4.2 Journey - language assistant (LLM)

HR opens Assist → types “Who is still not in today?” → retriever pulls attendance + person rows (and optional FAQ docs) → LLM answers in plain language with cited names/times → HR can open the person record. The assistant never marks attendance by itself; vision check-in does.

Chapter 5 - System Architecture & High-Level Design

5.1 End-to-end pipeline

Stage	Description
Data ingestion	Webcam / upload images (vision); user text prompts (language); browser GPS for punches; LFW offline for face eval; FAQ / wiki snippets for RAG
Vision engine	SCRFD detect → 5-pt align → ArcFace/FaceNet embed → FAISS cosine gallery; blink/texture anti-spoof
Language engine	Embed query → retrieve attendance/person/FAQ chunks → LLM generate grounded answer (RAG)
Fusion	Late fusion of modalities: vision decides identity/presence; language consumes structured events + text. Cross-modal handoff via shared person_id / attendance_id
Storage	PostgreSQL + PostGIS (prod); SQLite ok for pure unit tests; FAISS gallery; vector store for RAG chunks
Presentation	React + Vite SPA; FastAPI /api/v1

5.2 Architecture diagram (logical)

React SPA --JWT/images/text/GPS--> FastAPI (/api/v1)
 |-- /attendance/check-in --> Vision pipeline --> Gallery match --> Attendance row
 |-- /assist/chat -----> Retriever + LLM (RAG) --> Grounded answer
 |-- /schedule/sites -----> PostGIS geography CRUD + ST_DWithin gate
 PostgreSQL+PostGIS | FAISS face gallery | Chunk/vector index (RAG)

Chapter 6 - Technical Stack & Infrastructure

Component	Chosen	Rationale
Frontend / UI	React, Vite, TypeScript, Tailwind	SPA for dashboard + chat; fast iteration
Backend API	FastAPI, Uvicorn, SQLAlchemy	Same language as CV; OpenAPI; uploads
ML / Vision	OpenCV, SCRFD, ArcFace ONNX / FaceNet	Published encoders; CPU ONNX Runtime
ML / Language	Sentence embeddings + LLM API (or local SLM)	RAG over attendance + FAQ; tool-calling for SQL
Database	PostgreSQL + PostGIS	Required geospatial types/queries; org data

Component	Chosen	Rationale
Vector / search	FAISS (faces); pgvector or FAISS (text chunks)	Face cosine + RAG retrieval
Deployment	Docker Compose; Render/Railway/VM	API + web + Postgres/PostGIS

Assigned core stack remains Python + OpenCV + React + FastAPI. PostGIS is mandatory on the production database path (mentor / reference stack). Local SQLite may run UI tests without geo; any geofence acceptance test runs against Postgres+PostGIS.

Chapter 7 - Functional Requirements

ID	Requirement
FR-01	Platform admin creates organisations (slug, type) and can impersonate org context
FR-02	Org users login with organization code + username + password (JWT)
FR-03	Admin/HR CRUD people; enroll 1-5 face photos; deactivate removes from gallery
FR-04	Employee/kiosk check-in with webcam; present only if similarity $\geq \tau$
FR-05	Optional blink challenge rejects still-photo bursts (vision anti-spoof, not modality #2)
FR-06	When geofence on, check-in must send lat/lng; PostGIS ST_DWithin must pass
FR-07	Today dashboard: present / late / not in / waiting for photo
FR-08	CSV export of attendance for a date range
FR-09	LLM Assist: natural-language Q&A over attendance + people + FAQ with citations
FR-10	Sites CRUD with PostGIS geography point + radius_m
FR-11	Health endpoint reports encoder, gallery size, thresholds
FR-12	Role gates: employee cannot list all attendance; HR can

Chapter 8 - Non-Functional Requirements

- Performance: dashboard < 2 s; face inference < 1 s CPU; LLM first token < 5 s typical.
- Scalability: one org / campus (hundreds of identities); few concurrent check-ins.
- Security & privacy: hashed passwords; JWT expiry; HTTPS; biometric faces not public; LLM sees only org-scoped retrieved rows.
- Reliability: reject bad frames; 503 if encoder down; geofence fail-closed when required.
- Observability: access events + spoof/liveness logs; assist queries logged without raw secrets.

Chapter 9 - Data Strategy & Full Database Schema

9.1 Data sources

- Live enrollment photos and check-in frames (org biometric data).
- Punch GPS coordinates validated in PostGIS.
- LFW for offline face TAR@FAR (not mixed into live gallery).
- Org FAQ / policy snippets + generated attendance summaries for RAG (WikiQA-style lab set optional).

9.2 Core relational schema (PostgreSQL)

Table	Key columns
organization	id, name, slug UNIQUE, tz, work_start/end, geo_lat, geo_lng, geo_radius_m, geofence_on_self_punch, org_type, kernel, labels...
platform_user	id, username UNIQUE, hashed_password, is_admin, role
user	id, org_id, username, hashed_password, role, is_admin, person_id UNIQUE; UNIQUE(org_id, username)
person	id, org_id, name, employee_id, email, department, is_active, deactivated_at; UNIQUE(org_id, employee_id)
site	id, org_id, name, lat, lng, radius_m, is_default, geom geography(Point,4326) generated/stored for PostGIS
faceembedding	id, person_id, encoder, dim, vector (float[]/bytea), quality_json
attendance	id, org_id, person_id, user_id, site_id, decision, similarity, encoder, source, late, latitude, longitude, punch_geom geography(Point,4326), checked_out_at, created_at
accessevent	id, org_id, kind, person_id, decision, similarity, detail, created_at
holiday	id, org_id, day, name
spoofalert	id, org_id, actor_user_id, person_id, reason, image, created_at
assist_chunk	id, org_id, source_type, source_id, text, embedding vector - RAG chunks
assist_message	id, org_id, user_id, role, content, citations_json, created_at

9.3 PostGIS DDL (excerpt)

```
CREATE EXTENSION IF NOT EXISTS postgis;
ALTER TABLE site ADD COLUMN geom geography(Point,4326);
UPDATE site SET geom = ST_SetSRID(ST_MakePoint(lng, lat),4326)::geography;
CREATE INDEX site_geom_gix ON site USING GIST (geom);
-- inside fence if: ST_DWithin(site.geom, punch_geom, radius_m)
```

9.4 Privacy & versioning

Faces are biometric-class data: org-scoped storage; deactivate deletes gallery vectors. Code + PRD on GitHub; weights via download scripts; evaluation.json versioned; DVC/Git LFS optional for large assets.

Chapter 10 - API Specification

Base URL: /api/v1. Auth: Authorization: Bearer <JWT>. Platform admin may send X-Org-Id.

10.1 Auth & identity

Method	Path	Purpose
GET	/auth/config	Public flags, org labels, geofence, default site
POST	/auth/login	Body: username, password, org?; → access_token
POST	/auth/register	Employee signup when allowed
GET	/auth/me	User + linked person + org

Method	Path	Purpose
POST	/auth/password	Change password

10.2 People & enrollment

Method	Path	Purpose
GET/POST	/persons	List / create
GET/PATCH	/persons/{id}	Read / update
DELETE	/persons/{id}	Soft deactivate
POST	/persons/{id}/enroll	multipart images → embeddings
POST	/persons/{id}/login	Create linked user login
GET	/persons/{id}/timeline	Attendance + events

10.3 Attendance & geofence punches

Method	Path	Purpose
POST	/attendance/check-in	image + lat/lng + challenge_id? → present/reject
POST	/attendance/check-out	Optional face verify
POST	/attendance/kiosk	Staff kiosk face or PIN
GET	/attendance/today	Caller's today status
GET	/attendance	HR list (from/to/filters)
GET	/attendance/export	CSV download
POST	/attendance/manual	HR override

Check-in error codes: no_face, no_match, liveness_failed, location_required, outside_geofence.

10.4 Sites, assist (LLM), vision lab

Method	Path	Purpose
GET/POST	/schedule/sites	List / create PostGIS sites
PATCH/DELETE	/schedule/sites/{id}	Update / delete site
POST	/assist/chat	Body: {message} → {answer, citations[]}
GET	/assist/history	Prior assist messages
POST	/verify	Two-image similarity (lab)
POST	/identify	1:N identify
POST	/liveness/challenge/check	Blink challenge session
GET	/ops/summary	Today board aggregates
GET	/health	Encoder, gallery, thresholds

10.5 Example payloads

POST /assist/chat request: {"message": "Who is late today?"}

response: {"answer": "Rohan (EMP-12) checked in at 09:42 (late).", "citations": [{"type": "attendance", "id": Cul}]}

Chapter 11 - Track D: Multimodal AI (Vision + Language)

Mentor correction (v1.1): Pairing a static face crop with a blink video is not two modalities - both are vision. The required second modality is language / text via an LLM (audio may be added later as speech-to-text into the same text channel).

11.1 Multimodal data fusion

Stream	Modality	Role
A - Vision	RGB face image (+ optional blink burst)	Who is present? Anti-spoof inside vision
B - Language	Natural language text (+ optional STT audio→text)	What does attendance mean? Org Q&A

Fusion strategy: Late / decision-level fusion with shared identifiers. Vision writes attendance rows keyed by person_id. The language stack retrieves those rows (and FAQ chunks), then an LLM generates an answer. We do not claim early pixel-text cross-attention transformers for v1; RAG + structured tools is the track-appropriate fusion.

11.2 Model architecture

- Vision: SCRFD → align → ArcFace/FaceNet → 512-D → FAISS cosine; blink/EAR + freeze cues.
- Language: query embedder → top-k chunk / SQL tool retrieval → LLM (API or local SLM) with citation prompt template.
- Reference: Deng et al., ArcFace (CVPR 2019); Schroff et al., FaceNet (CVPR 2015); Lewis et al., RAG for Knowledge-Intensive NLP (2020).

11.3 Fine-tuning & prompt strategy

- Face encoder: prefer published weights; optional small FaceNet fine-tune with LFW identities excluded.
- LLM: prompt templates force “answer only from retrieved rows; say unknown if missing.”
- Optional PEFT/LoRA on a small SLM in Month 3 if API cost / offline demo requires it.

11.4 XAI, safety & latency

- Vision XAI: show similarity, match name, failure reason codes; lab verify cosine.
- Language XAI: show citations (attendance id / person id / FAQ chunk).
- Safety: no attendance mutation from chat; org isolation on retrieval; hallucination → “I don’t know”.
- Metrics: face TAR@FAR; assist grounded-answer rate; RAGAS-style faithfulness on a 50-Q lab set.
- Latency: exact face cosine for small N; cache FAQ embeddings; bound LLM context.

Chapter 12 - Geospatial Design (PostGIS)

The reference stack requires a PostGIS geospatial component. Attendance punches are not trusted without a spatial predicate against organisation sites.

12.1 Spatial model

- Each site stores lat, lng, radius_m and geom geography(Point, 4326).
- Each punch stores latitude, longitude and punch_geom.
- Gate: ST_DWithin(site.geom, punch_geom, radius_m) (metre distances on geography).
- Org-level office fence uses the same pattern when no multi-site list exists.

12.2 Behavioural rules

Condition	System behaviour
geofence_on_self_punch = false	GPS optional; no PostGIS reject
geofence on, GPS missing	400 location_required
Point outside all fenced sites	403 outside_geofence
Kiosk on trusted device	Policy may skip GPS; still logged

12.3 Why not Haversine-only

A Python Haversine helper may remain as a fallback for SQLite unit tests, but production deployments must enable PostGIS and execute spatial checks in SQL so the reference stack and indexing (GiST) are real, reviewable artefacts.

Chapter 13 - UI/UX & Presentation Layer

- Login: org code + username/password; catchy org-focused hero copy.
- Employee: Check in (camera oval), My attendance calendar.
- HR: Today board, People, Enroll, Sites/geofence, Reports/CSV, Assist (LLM chat).
- Lab: Verify / Identify / Liveness for operators.
- Default theme: light; optional dark toggle. Responsive layout. Dashboard load < 2 s.

Chapter 14 - Testing, QA & Validation

- Unit/integration: Pytest for auth, enroll, check-in shape, CSV, role 403s, assist citation schema; Vitest for login/dashboard.
- Model: LFW View 2 TAR@FAR; still-photo fails blink; RAG faithfulness on fixed FAQ set.
- PostGIS: Dockerized Postgres+PostGIS tests - inside/outside radius fixtures.
- Edge cases: no face, multi-face, unknown visitor, deactivated person, GPS denied, LLM with empty retrieval → honest unknown.

Chapter 15 - Deployment, CI/CD & Milestones

15.1 Engineering practices

- GitHub: main / development / feature/* ; PRs; tags before vivas.
- Docker Compose: api, web, postgres:postgis image.
- CI: GitHub Actions - pytest (+ PostGIS service) + frontend build.
- Live URL + demo video by 25 November 2026.

15.2 Milestone roadmap

Phase	Deliverables	Deadline
PRD approval	This v1.1 PRD reviewed/approved	30 Aug - 3 Sept 2026
GitHub base	Repo layout, README, Compose with PostGIS	6 September 2026
Viva 1	Face pipeline + TAR@FAR; JWT; schema; PostGIS sites DDL; assist stub	15-22 September 2026
Viva 2	Enroll/check-in UI; geofence gate; LLM RAG chat; CSV; API tests	15-22 October 2026
Viva 3	E2E polish; grounded-answer eval; Docker deploy; model card	15-22 November 2026
Final	Live link + demo video	25 November 2026
External viva	Defence + live demo	15-22 December 2026

Chapter 16 - Risk Management & Requirements Traceability Matrix

16.1 Risks

Risk	Severity	Mitigation
CPU / no GPU	High	ONNX ArcFace; Colab only if fine-tuning
LLM cost / rate limits	Medium	Cache retrieval; local SLM fallback; tight prompts
LLM hallucination	High	Citations required; refuse when retrieval empty
False reject / accept	High	LFW τ ; re-enroll; blink anti-spoof
PostGIS ops complexity	Medium	Compose image; fallback tests documented
Scope creep	High	Freeze FRs in Ch.7; stretch after Viva 2

16.2 Requirements Traceability Matrix (RTM)

Maps each functional requirement to design chapters, primary APIs/tables, and viva proof.

Req	Design ch.	Primary artefact	Verification
FR-01	4, 9, 10	organization, /orgs	Viva 1 - create org
FR-02	4, 10	POST /auth/login	Viva 1 - JWT org login
FR-03	7, 9, 10	person, faceembedding, /persons/*/enroll	Viva 2 - enroll UI
FR-04	5, 11	POST /attendance/check-in	Viva 2 - present decision
FR-05	11, 14	/liveness/*	Still-photo fails test
FR-06	12, 9	PostGIS ST_DWithin	Inside/outside fixture
FR-07	13, 10	GET /ops/summary	Dashboard < 2 s
FR-08	7, 10	GET /attendance/export	CSV download
FR-09	11, 10	POST /assist/chat + assist_* tables	Grounded FAQ $\geq 90\%$

Req	Design ch.	Primary artefact	Verification
FR-10	12, 10	/schedule/sites + site.geom	CRUD + GiST index
FR-11	10	GET /health	CI smoke
FR-12	8, 10	role deps	403 pytest

Appendix A - Roles (summary)

Role	Capabilities
Platform admin	All orgs; no org code; X-Org-Id
Org admin / HR	People, enroll, sites, reports, Assist, Today
Employee	Check-in, own attendance
Operator	Kiosk + vision lab (no people admin)

Appendix B - Model card outline (Month 3)

- Vision input: aligned RGB crop → 512-D unit embedding; decision cosine $\geq \tau$ @ FAR=1e-3 + anti-spoof.
- Language input: user text → retrieved org rows/chunks → LLM answer with citations.
- Limitations: lighting/pose/masks; LLM may abstain; geofence needs GPS permission.
- Ethics: workplace/school notice; delete embeddings on exit; org data isolation.

End of comprehensive PRD v1.1 - Face Attendance System (Multimodal AI: Vision + Language) - Polaris School of Technology OJT.
 Prepared for resubmit to mentor Soumen Mukherjee.